Research Center E. Piaggio
University of Pisa

Master's Thesis and Project Proposal

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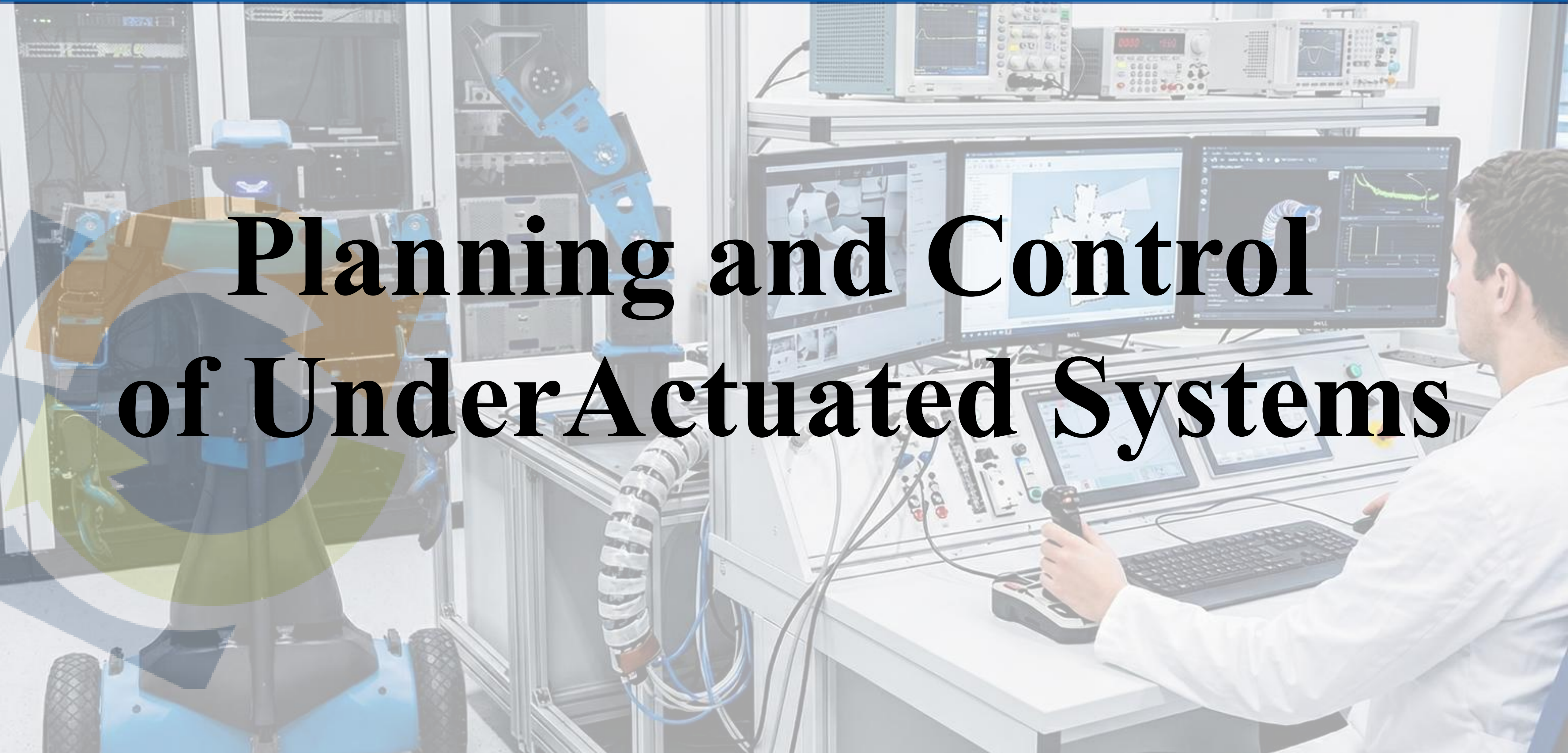
Planning and Control of UnderActuated Systems



Dynamic Adaptive Control of VSA for Object Manipulation

Overview

Adaptive control is commonly used for compensating for the uncertainties in the robot model. During the grasping task, the uncertainty arises about the grasped object's inertial parameters. The main challenge of this thesis is the formalization of the dynamic adaptive control for the Variable Stiffness Actuators, including both the motor and the link dynamics.

Goal

The goal of this thesis is investigate the best solution for this problem, studying the different solutions present in literature (direct, indirect, and composite adaptive control), understanding how we can fit the strategies for an underactuated arm. The student will implement it in the real robot, using ROS.

Expected Results

The expected outcomes are: i) the student will go into detail about the adaptive control strategies and the Variable Stiffness Actuators; ii) generalization of the chosen strategy for the underactuated arm; iii) experimental validation with the Alter-Ego robot.

Tools and Programs

ROS, Gazebo, C++/Python, Alter-Ego

Robot Redundancy Handling to Compensate Unstable Base Oscillation

Overview

In various robotic applications, the robot may operate on a base that is subject to oscillations or external disturbances, such as a mobile platform or an unstable support surface. These unwanted base motions negatively affect the stability of the manipulator and the precision of the task execution. This thesis focuses on exploiting robot redundancy to counteract these oscillations, redistributing joint motions in a way that maintains the desired end-effector behavior even when the base is not stationary.

Goal

The goal of this thesis is to investigate and develop redundancy-based control strategies capable of:
compensating for base movements when these are known in advance or sensed in real time;
reacting effectively to unexpected or unknown oscillations; integrating the proposed strategy within the robot controller and validating its performance through simulation or experiments.

Expected Results

The expected outcomes include:
i) an in-depth literature review on redundancy resolution methods for robots experiencing base motion; ii) the design and implementation of at least one redundancy strategy addressing both known and unknown oscillations; iii) experimental or simulation-based validation of the proposed approach

Tools and Programs

ROS, Gazebo, C++/Python, Alter-Ego

Sensorless Stiffness Estimation on VSA

Overview

Variable Stiffness Actuators (VSA) allow robots to modulate compliance, improving safety and interaction performance. However, accurately estimating the actuator stiffness often requires dedicated sensors, which increase system complexity, cost, and fragility. This thesis focuses on sensorless stiffness estimation, i.e., inferring the stiffness from available motor signals and actuator dynamics without relying on additional hardware. The challenge lies in designing an estimation method capable of dealing with nonlinear VSA models and real-world uncertainties.

Goal

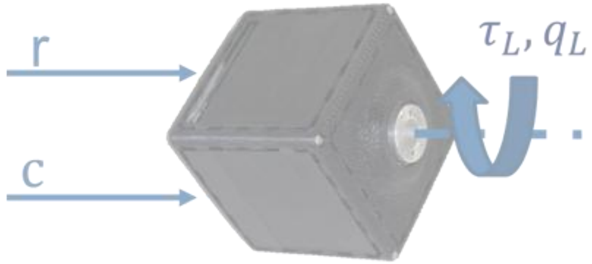
The goal of the thesis is to study existing sensorless estimation techniques and implement one of the most promising approaches for a selected VSA mechanism. The work involves:

•reviewing the state of the art on VSA modeling and sensorless stiffness estimation;
•selecting one estimation algorithm suitable for the considered actuator;
•implementing the approach and adapting it to the real system's behavior.

Expected Results

The expected outcomes are:

i) a complete literature review on Variable Stiffness Actuators and sensorless stiffness estimation approaches;
ii) implementation of at least one selected estimation method;
iii) experimental validation of the approach on the real actuator, assessing accuracy and robustness.



Tools and Programs

C++/Python, ROS, experimental setup with the VSA platform

VSA identification with Data-Driven Approaches

Overview

Variable Stiffness Actuators (VSA) exhibit complex, nonlinear behaviors that are often difficult to model analytically. Data-driven identification techniques offer a promising alternative by learning the actuator dynamics directly from experimental data rather than relying on simplified physical models.

This project focuses on exploring and comparing modern data-driven approaches for identifying VSA behavior, with the goal of achieving accurate and robust models that can be integrated into control pipelines for humanoid and human-robot interaction applications.

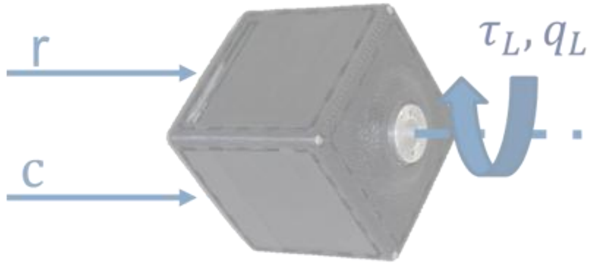
Goal

The goal of the project is to study, implement, and evaluate data-driven identification techniques for VSA systems. The work includes:

•conducting a literature review on data-driven identification methods;
•implementing 2–3 identification approaches (e.g., regression models, neural networks, nonlinear system identification);
•comparing the learned models with the real actuator behavior through experiments.

Expected Results

The expected outcomes include:
i) a comprehensive literature review on VSA modeling and data-driven identification methods;
ii) implementation of multiple identification approaches and analysis of their performance;
iii) comparison with real-system data to evaluate accuracy, generalization, and limitations.

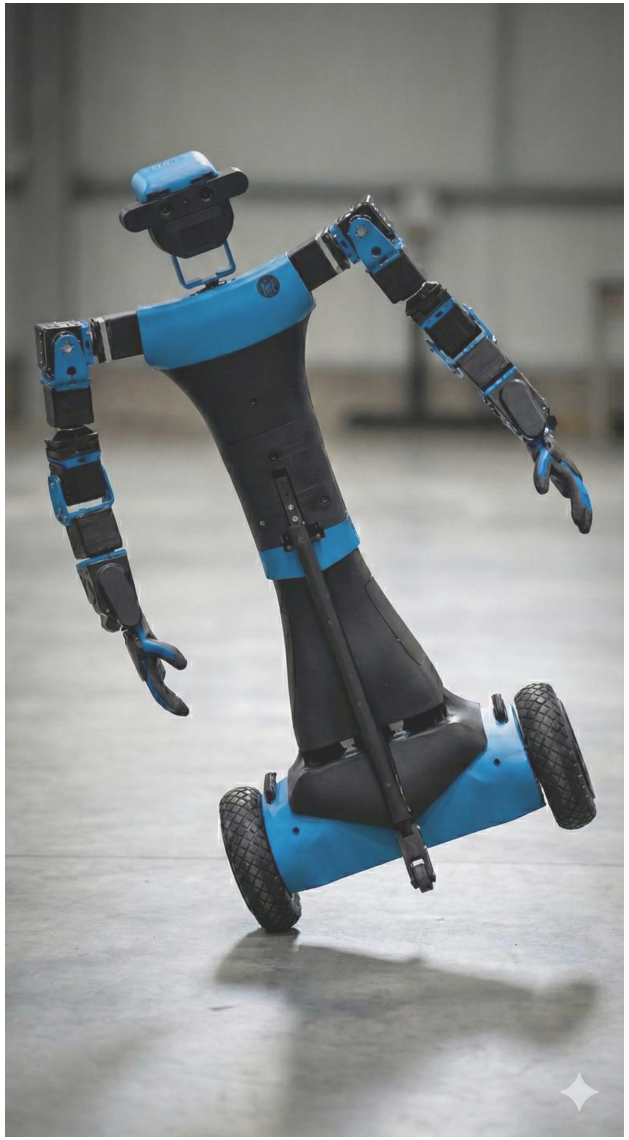


Tools and Programs

Python/C++, ROS, Neural Networks frameworks, experimental VSA setup

‘External’ Tutors: Simone Tolomei, Yuri De Santis

Whole-Body Balancing Control with a single wheel



Overview

Balancing a humanoid-like robot on a single wheel represents a highly dynamic and inherently unstable control problem. The Alter-Ego platform, equipped with compliant actuation and whole-body control capabilities, provides an ideal testbed for investigating balance control under strong dynamic constraints.

This thesis aims to study, model, and control the Alter-Ego robot when performing balance on a wheel, addressing the challenges posed by the underactuated and highly nonlinear dynamics of the system.

Goal

The goal of the thesis is to design and validate a complete framework for controlling the robot while balancing on a wheel. The work includes:

performing a literature review on dynamic balancing and wheeled inverted-pendulum systems;
developing a dynamic model of the Alter-Ego system in the wheel-balancing configuration;
designing a dynamic control strategy capable of maintaining balance;
validating the control approach through simulations in Gazebo and, if feasible, experiments on the real robot.

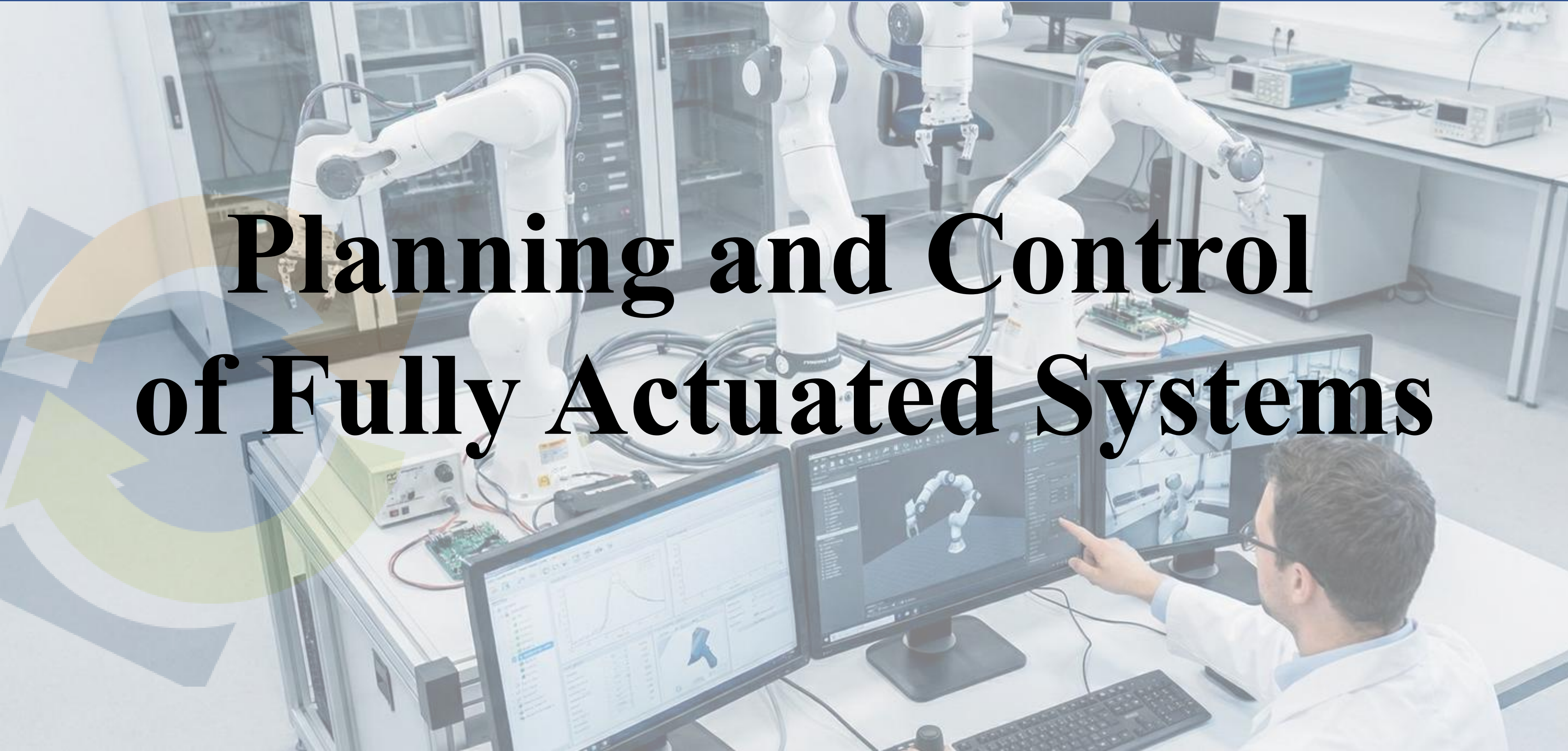
Expected Results

The expected outcomes include:
i) a detailed review of the literature on dynamic balance control for underactuated wheeled systems;
ii) a dynamic model capturing the relevant behavior of the Alter-Ego robot on a wheel;
iii) a control strategy for balancing, tested in simulation and possibly experimentally validated.

Tools and Programs

Python/C++, ROS, Alter-Ego

Planning and Control of Fully Actuated Systems



Reverse Priority Algorithm with Unilateral Constraints by Yolo Segmentation in Franka Emika Panda and Active Wrist Platform

Overview

This project focuses on the implementation and integration of a reverse-priority control framework for the Franka Emika Panda robot equipped with a soft wrist. The control problem involves managing unilateral constraints while enabling human-like motion primitives.

The work additionally incorporates a perception module based on YOLO, whose output must be integrated into the reverse-priority controller to enable object-aware and task-oriented robot behavior. The final objective is to design a fully integrated pipeline that performs robustly in real-world scenarios.

Goal

The goal of the project is to design, implement, and experimentally validate a control pipeline that combines perception and task-priority strategies. The work includes:
implementing a YOLO-based perception algorithm;
integrating YOLO outputs into a reverse-priority control framework;
ensuring proper handling of unilateral constraints;
conducting experiments with the Franka Emika Panda to validate the approach.

Expected Results

The expected outcomes include:
i) a working implementation of the YOLO perception module;
ii) a reverse-priority control framework integrated with perception;
iii) experimental validation on the Franka robot demonstrating feasibility and robustness.

Tools and Programs

Python/C++, ROS, Franka Emika Panda, Yolo (or other Computer Vision tools)

'External' Tutors: Giorgio Simonini, Marco Baracca

Project / Master Thesis

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Navigation and Grasping

Whole-Body Planning for Cart Pushing Task

Overview

This project focuses on planning and controlling the Alter-Ego robot while interacting physically with a cart. The task involves modeling the system as an unicycle with a front trailer, extending it by including two arm connections, and defining suitable control objectives for coordinated cart-pushing. The project aims to combine locomotion, manipulation, and physical interaction within a unified control framework.

Goal

The goal of the project is to develop a planning and control strategy that enables Alter-Ego to push and guide a cart effectively. The work includes:

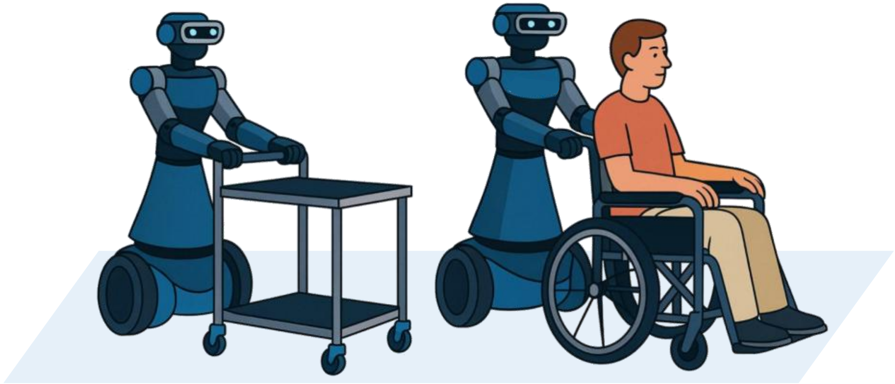
•modeling the system as an unicycle with a front trailer;
•extending the model by adding the two arm connections;
•defining control goals (arm positions or relative positioning between robot and cart);
•designing a new control strategy that achieves stable and coordinated cart-pushing;
•performing experimental validation if feasible.

Expected Results

The expected outcomes include:
i) a complete model of the robot–cart interaction, including arms and trailer dynamics;
ii) a planning and control strategy enabling coordinated pushing;
iii) experimental validation of the proposed approach.

Tools and Programs

Matlab, Python/C++, ROS, Alter-Ego



Reinforcement Learning to Push a Cart with the Alter-Ego robot

Overview

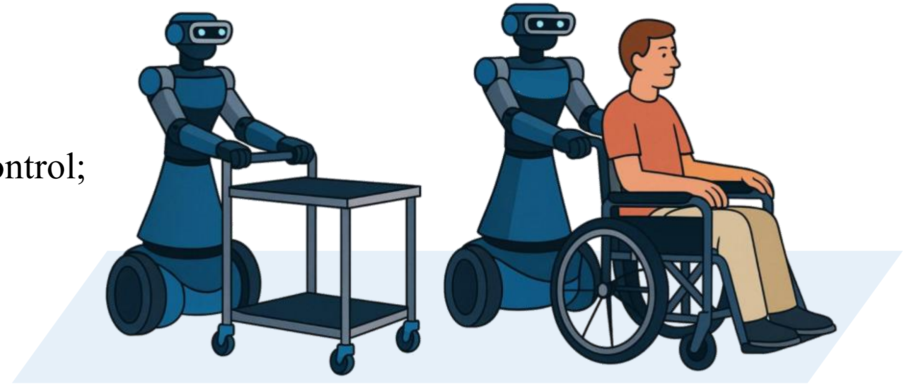
This thesis focuses on applying Reinforcement Learning (RL) to enable the Alter-Ego humanoid platform to interact physically with a cart and perform coordinated pushing behaviors.
The task requires modeling the system as a unicycle with a front trailer and incorporating the effect of the two arm contacts. The goal is to train an RL policy capable of generating stable, coordinated whole-body actions that achieve effective cart-pushing.
The work includes studying relevant RL algorithms and adapting them to the Alter-Ego robot simulated inside IsaacLab.

Goal

The goal of this thesis is to develop and validate a Reinforcement Learning framework for cart-pushing with the Alter-Ego robot. The work includes:
•conducting a literature review on Reinforcement Learning for locomotion and interaction tasks;
•implementing the Alter-Ego robot model inside IsaacLab;
•training an RL model to perform the cart-pushing task;
•validating the learned policy in simulation and, if feasible, on the real platform.

Expected Results

The expected outcomes include:
i) a complete literature review on Reinforcement Learning and its applications in humanoid control;
ii) a working RL model trained to perform the cart-pushing task;
iii) experimental validation in simulation and possible real-robot testing.



Tools and Programs

Python/C++, ROS, Alter-Ego, IsaacLab

‘External’ Tutors: Simone Tolomei, Yuri De Santis

Master Thesis

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NaviGrasping: Multitask Planning for Navigation and Grasping

Overview

This thesis focuses on enabling a robot to navigate in the environment while simultaneously performing grasping actions. The project lies at the intersection of navigation, whole-body control, and manipulation, requiring a unified framework capable of coordinating locomotion and grasp execution.
The work includes reviewing the literature on multitask planning, whole-body grasping controllers, and navigation-manipulation integration, with the final aim of developing a system that performs grasping while moving.

Goal

The goal of the thesis is to design and validate a whole-body control strategy that enables grasping during navigation.
The work includes:
conducting a literature review on multitask navigation and grasping approaches;
designing a whole-body control strategy for grasping while the robot navigates;
implementing the method in simulation or on a real robot;
performing experimental validation.

Expected Results

The expected outcomes include:
i) a full literature review on navigation and grasping multitask planning;
ii) a whole-body control strategy enabling grasping during navigation;
iii) experimental validation of the proposed approach.

Tools and Programs

Python/C++, ROS, Alter-Ego

Master Thesis

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Real-Time Planning of a Navigation Task Using AI

Overview

This thesis focuses on developing an AI-based navigation system capable of interpreting vocal commands, generating a navigation plan, and executing it autonomously. The project integrates speech-to-text algorithms, planning tree strategies, and robot navigation, with the goal of enabling intuitive and intelligent human-robot interaction.

Goal

The goal of this thesis is to design and validate a complete AI-assisted navigation framework. The work includes:

- conducting a literature review;
- implementing a speech-to-text algorithm (e.g., using the ChatGPT API);
- generating a planning tree based on vocal requests;
- designing a navigation strategy that estimates and follows the appropriate node in the planning tree;
- performing experimental validation on the robot.

Expected Results

The expected outcomes include:

i) an effective speech-to-text interface for navigation commands;
ii) a planning tree strategy capable of translating verbal instructions into navigable goals;
iii) a fully implemented and validated navigation system using AI-assisted planning.

Tools and Programs

Python/C++, ROS, Alter-Ego, AI tools

Master Thesis

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Visual SLAM with RealSense for 3D Mapping and Grasping planning

Overview

This project focuses on implementing a Visual SLAM algorithm using an Intel RealSense camera to enable real-time 3D mapping for robot perception and manipulation.
The aim is to build a robust SLAM pipeline capable of generating reliable environmental maps and depth information that can be integrated into the Alter-Ego robot’s MoveIt planner.
The system will ultimately support grasping and object-interaction tasks by providing accurate spatial information in dynamic or semi-structured environments.

Goal

The goal of this project is to develop a complete SLAM-based perception and planning pipeline. The work includes:
•performing a literature review on Visual SLAM algorithms;
•implementing one selected SLAM algorithm and validating it with a RealSense camera;
•integrating the SLAM-generated information with the Alter-Ego MoveIt planner;
•testing the resulting system in simulation and/or real scenarios.

Expected Results

The expected outcomes include:
i) a complete literature review on Visual SLAM methods;
ii) a working implementation of a Visual SLAM algorithm validated with RealSense;
iii) integration of the mapping output with the Alter-Ego planning framework;
iv) demonstration of grasping or object-interaction tasks using SLAM-based perception

Tools and Programs

Python/C++, ROS, Alter-Ego, Computer Vision Tools, MoveIt

Visual servoing to correct the Alter-Ego pose in the grasping task

Overview

This task focuses on using visual servoing to correct in real time the position of the Alter-Ego robot's base during a grasping operation. The goal is to reduce positioning errors caused by imperfect perception, slippage, or small environmental changes.

Goal

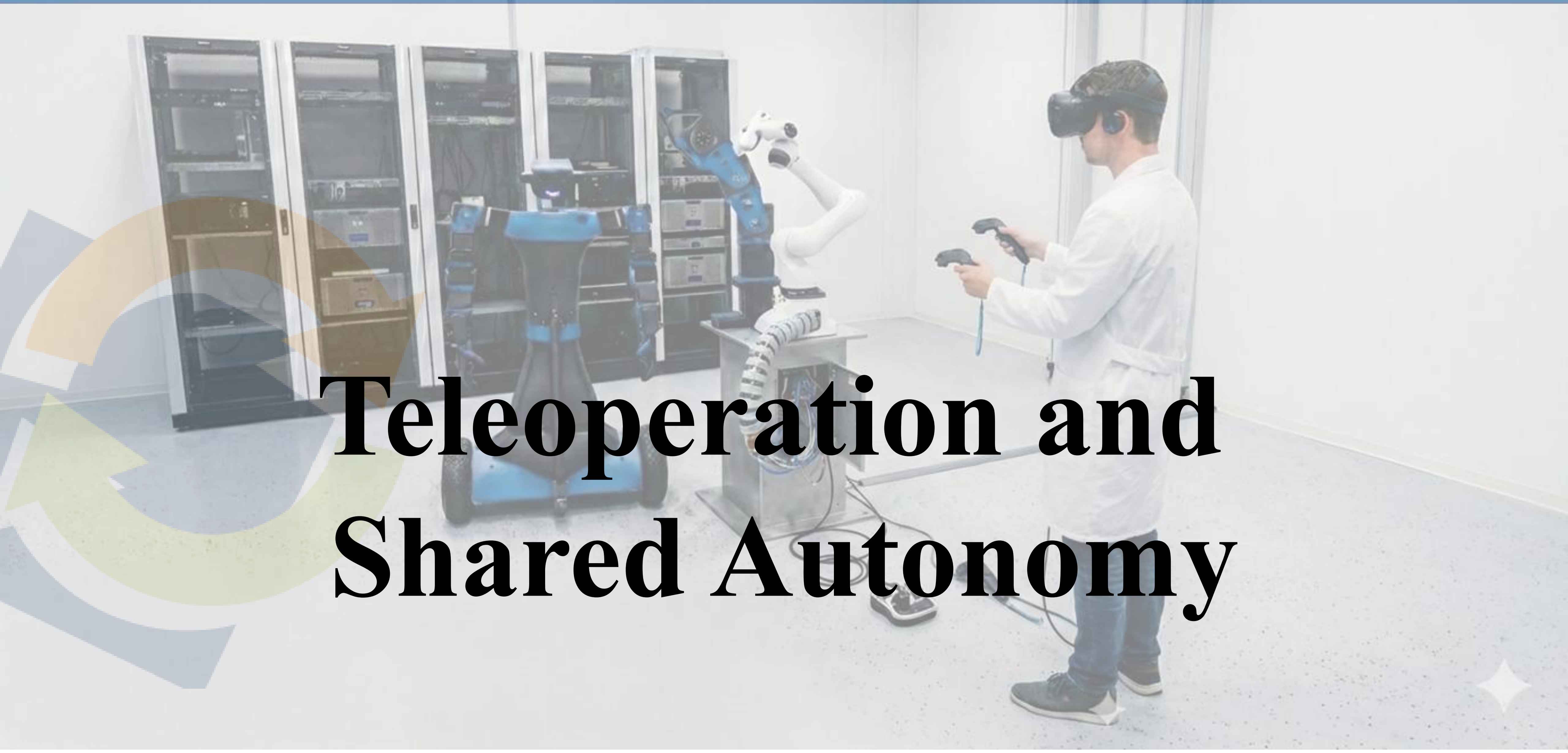
Develop a visual-servoing pipeline capable of estimating the deviation between the current base position and the ideal grasping configuration. Use visual feedback from the robot's camera to adjust the base trajectory. Integrate the servoing module into Alter-Ego's motion-planning architecture.

Expected Results

A fully working visual-servoing system integrated with Alter-Ego's locomotion. A measurable reduction in base-positioning error during grasping tasks. Potential integration with SLAM-based perception or motion-planning modules.

Tools and Programs

Python/C++, ROS, Alter-Ego



Teleoperation and Shared Autonomy

Collision Avoidance with Shared Autonomy with Teleoperation interface strategy

Overview

This project focuses on developing a shared-autonomy framework that enables a robot to assist the human operator by preventing collisions during teleoperation. The system must distinguish when an object should be considered an obstacle and when it is actually the target of the task.

Goal

Perform a literature review on collision avoidance and shared autonomy. Design a shared-autonomy algorithm capable of modulating robot behavior based on the operator's intent. Develop an interface for managing the distinction between obstacles and task-relevant objects.

Expected Results

Validation of the algorithm in simulation. Experimental validation on the real system. Design a useful interface for the teleoperator to adjust the sharing factor with the joystick.

Tools and Programs

Python/C++, ROS, Alter-Ego

Hand-Motion Teleoperation with Filter for Parkinson’s Patients

Overview

This project focuses on developing a hand-motion teleoperation system specifically adapted for Parkinson’s patients, where motion irregularities and tremors require a robust filtering strategy. It involves studying teleoperation, shared-autonomy principles, and human–robot interaction to enable stable and intuitive control.

Goal

Perform a literature review on teleoperation strategies, human-motion filtering, and assistive-robotic interfaces. Design a hand-motion teleoperation algorithm including a dedicated filtering method for Parkinson-related tremors. Develop the implementation in Python (as required in the task description).

Expected Results

Validation of the filtering and teleoperation algorithm in simulation. Experimental validation on the real robotic system. Demonstration of improved control stability and usability for subjects with motion irregularities.

Tools and Programs

Python/C++, ROS, Alter-Ego, Experimental Validation with Healthy or not Subjects

Interface with the Eye-Tracking tool and the Webapp for the Motor Impairment Patients

Overview

This project focuses on developing an interface using an eye-tracking tool integrated with a web application to support patients with motor impairments. The task involves studying teleoperation, shared-autonomy principles, and human–robot interaction to create an accessible and intuitive control system.

Goal

Perform a literature review on teleoperation strategies, human-motion filtering, and assistive-robotic interfaces. Design a teleoperation interface based on eye-tracking input tailored for motor-impaired users. Implement the control and filtering algorithms in Python.

Expected Results

Validation of the filtering and interaction algorithms in simulation. Experimental validation on the real robotic system. Demonstration of improved interaction stability and usability for users with motor impairments.

Tools and Programs

Python/C++, ROS, Alter-Ego, WebApp, Experimental Validation with Healthy or not Subjects

Facial Expression Recognition and Reproduction in Ego for Emotion-Aware Task

Overview

This project focuses on facial expression recognition using neural networks and the reproduction of the detected expression on the Ego robot.
The work lies within the domains of Human-Robot Interaction and Assistive Robotics, and it includes creating a demo where Ego autonomously reacts to the user (e.g., welcoming behaviors).

Goal

Conduct a literature review on facial expression recognition and affective human-robot interaction. Implement a facial expression recognition algorithm using neural networks in Python. Reproduce the recognized expression on the Ego robot in autonomous mode. Develop a demo showcasing the full perception-to-actuation pipeline.

Expected Results

A functional neural-network-based facial expression recognition module. An integrated system where Ego reproduces expressions based on real-time visual input. A demonstration scenario (e.g., Ego welcoming the user) showing emotion-aware robot behavior.

Tools and Programs

Python/C++, ROS, Alter-Ego

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